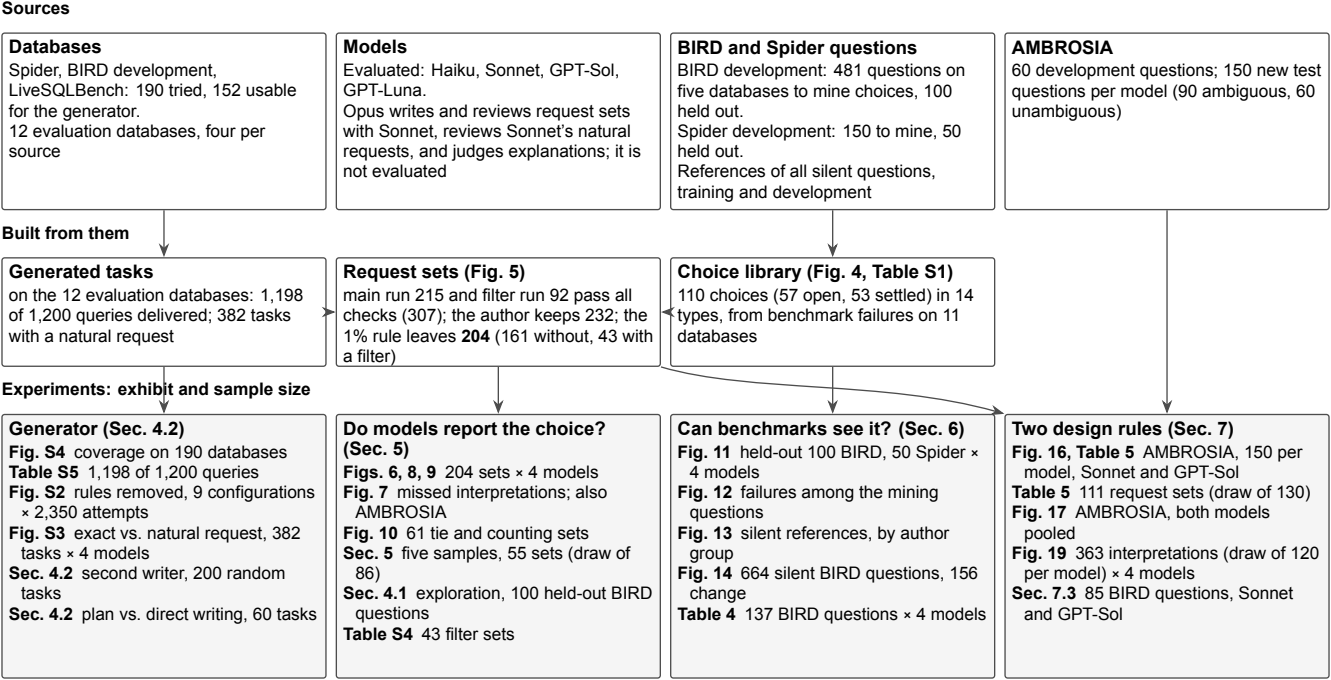
Sources
Databases
Spider, BIRD development, LiveSQLBench: 190 tried, 152 usable for the generator.
12 evaluation databases, four per source
Models
Evaluated: Haiku, Sonnet, GPT-Sol, GPT-Luna.
Opus writes and reviews request sets with Sonnet, reviews Sonnet's natural requests, and judges explanations; it is not evaluated
BIRD and Spider questions
BIRD development: 481 questions on five databases to mine choices, 100 held out.
Spider development: 150 to mine, 50 held out.
References of all silent questions, training and development
AMBROSIA
60 development questions; 150 new test questions per model (90 ambiguous, 60 unambiguous)
Built from them
Generated tasks
on the 12 evaluation databases: 1,198 of 1,200 queries delivered; 382 tasks with a natural request
Request sets (Fig. 5)
main run 215 and filter run 92 pass all checks (307); the author keeps 232; the 1% rule leaves 204 (161 without, 43 with a filter)
Choice library (Fig. 4, Table S1)
110 choices (57 open, 53 settled) in 14 types, from benchmark failures on 11 databases
Experiments: exhibit and sample size
Generator (Sec. 4.2)
Fig. S4 coverage on 190 databases
Table S5 1,198 of 1,200 queries
Fig. S2 rules removed, 9 configurations × 2,350 attempts
Fig. S3 exact vs. natural request, 382 tasks × 4 models
Sec. 4.2 second writer, 200 random tasks
Sec. 4.2 plan vs. direct writing, 60 tasks
Do models report the choice? (Sec. 5)
Figs. 6, 8, 9 204 sets × 4 models
Fig. 7 missed interpretations; also AMBROSIA
Fig. 10 61 tie and counting sets
Sec. 5 five samples, 55 sets (draw of 86)
Sec. 4.1 exploration, 100 held-out BIRD questions
Table S4 43 filter sets
Can benchmarks see it? (Sec. 6)
Fig. 11 held-out 100 BIRD, 50 Spider × 4 models
Fig. 12 failures among the mining questions
Fig. 13 silent references, by author group
Fig. 14 664 silent BIRD questions, 156 change
Table 4 137 BIRD questions × 4 models
Two design rules (Sec. 7)
Fig. 16, Table 5 AMBROSIA, 150 per model, Sonnet and GPT-Sol
Table 5 111 request sets (draw of 130)
Fig. 17 AMBROSIA, both models pooled
Fig. 19 363 interpretations (draw of 120 per model) × 4 models
Sec. 7.3 85 BIRD questions, Sonnet and GPT-Sol